

OpenSky receiver: setup guide

A one-page plan for the Brockenhurst Community Action Group. Running a small ADS-B receiver earns us access to OpenSky's full historical flight archive, and records the low arrivals over the village into an independent public dataset. Prices are approximate (July 2026) and worth a quick check before buying.

Why we're doing it

- **Data access.** OpenSky grant historical database access to contributors. A receiver is the agreed route to the archive we asked for.
- **Our own record.** From the day it is switched on, it captures the flights over Brockenhurst into a public, timestamped dataset that nobody can dismiss as third-party.
- **Better local coverage.** ADS-B is line of sight, and coverage over the New Forest is thin. A receiver here improves it, which strengthens any comparison over time.

Shopping list

Item	What it is and why	Approx price
Raspberry Pi 4 (2GB)	The little computer that runs it. A Pi 3B+ is fine too.	£40 to £55
Official USB-C power supply	Steady power matters; cheap chargers cause dropouts.	£10
microSD card, 16 to 32GB	Holds the software image.	£8
RTL-SDR dongle	Receives the 1090 MHz signals. A FlightAware Pro Stick Plus has a built-in filter and is the easy choice.	£25 to £35
1090 MHz ADS-B antenna	The single biggest factor in how much you pick up. A proper 1090 antenna beats any stock one.	£20 to £40
Coax cable (if loft or roof mounted)	Connects antenna to dongle. Keep it short.	£10
Case (optional)	Keeps the Pi tidy.	£8

Realistic total: about £120 for a good setup, or nearer £70 if you go lean (a Pi Zero 2 W and a basic antenna).

Where to buy

Search these exact product names at a trusted retailer rather than the cheapest listing:

- Raspberry Pi and power supply: **The Pi Hut** (thepihut.com) or **Pimoroni**.
- RTL-SDR dongle: **FlightAware Pro Stick Plus**, or **RTL-SDR Blog V4** (rtl-sdr.com).
- 1090 MHz antenna: **FlightAware 1090 MHz ADS-B antenna**.

If a group member already runs a receiver for FlightAware, Flightradar24 or ADS-B Exchange, we may not need new hardware at all. OpenSky can usually be added as an extra feed on the same box. Ask around first.

Where to put it (the one decision that matters)

1090 MHz is line of sight, so **higher and clearer wins every time**.

- Best host: someone in or near Brockenhurst with an open view of the sky toward the **south and east**, the side the arrivals come in from.
- A loft or an upstairs window usually works. A roof or short mast is better.
- It needs mains power and wifi or ethernet, and runs 24/7. Power use is a few watts, so the electricity cost is negligible.

Sited well, it will record the exact low overflights this campaign is about.

Does it need to be under the flight path?

No. The receiver is a listening post, not a camera. Every aircraft broadcasts its own GPS position, so it logs exactly where each plane is no matter where the antenna sits. One receiver in Brockenhurst records the low arrivals over the whole area, because it hears their broadcasts from miles off.

The one thing that gets missed is aircraft that are both low and far away, which slip below the horizon behind trees and terrain. That is why sitting in the area where they fly low is ideal: a low plane passing overhead is the easiest signal of all to catch. So one well-sited receiver in the village covers the whole local corridor. There is no need to chase the exact line they fly.

Indoors, window, loft or outside?

It does not have to be outdoors. Reception improves in steps, so this is about how much you catch, not pass or fail:

- **Indoors in a room:** works, but walls and insulation weaken the signal. Still catches the low flights right overhead, which is our main goal.
- **By a window:** better and easy. A window only sees the way it faces, so point it at open sky to the south or east. No need to put it outside.
- **In the loft:** usually the best practical spot, elevated and out of the weather. **Important gotcha:** many modern roofs have a foil-backed membrane under the tiles that blocks the signal. If your roof has it, use a window instead. Older tile or plain-felt roofs are fine.
- **Outside on the roof or a mast:** best performance, but means weatherproofing and a cable run. Overkill unless someone is keen.

For our purposes a loft antenna (non-foil roof) or one by an upstairs south or east facing window is plenty. Quick loft check: if you can see daylight through gaps it will likely work; if you see a shiny foil membrane, go with a window.

Setup, step by step (for whoever hosts it)

The official instructions are at opensky-network.org/feed/raspberry. Follow those for the

exact download and commands. Here is what each stage means so it is not a mystery:

- **1. Flash the image.** Download OpenSky's Raspberry Pi image and write it to the SD card using Raspberry Pi Imager or Balena Etcher. This is the whole operating system plus their feeder software, ready to go.
- **2. Connect the hardware.** Plug the SDR dongle into a USB port, screw the antenna onto the dongle, insert the SD card, connect network and power. Place the antenna as high and clear as you can.
- **3. Register the receiver.** Log in to the OpenSky account, add the sensor, and enter the antenna's exact location: latitude, longitude and height above ground. Accuracy here directly affects data quality, so measure it properly.
- **4. Check it is feeding.** The OpenSky account page shows your sensor as online once data is flowing. Give it a few minutes.
- **5. Follow up.** Once it is confirmed feeding, reply to Martin (Ticket #835878) to request the historical (Trino) access.

After it is running

- Leave it powered on. The value grows the longer it collects.
- We will then have both the historical archive to analyse and a live local feed of our own.
- If it drops offline, it is almost always power or network. A reboot usually fixes it.

Any group member comfortable with a Raspberry Pi can do this in an afternoon. If nobody is, the guide is simple enough to follow, or a local enthusiast can help. Michael will coordinate the setup and the OpenSky side.